

Two-ness Clinics — The Receipt Test

Tagline: *Can you tell them apart later? Only if you kept the receipt.*

This document compiles the finalized, executable synthesis of the **Two-ness Clinics** into a single program that works across:

- **Tier 0:** museum / public pop-up (7–10 minutes)
- **Tier 1:** classroom / undergrad lab (60–120 minutes)
- **Tier 2:** research-grade experiments and papers

It also includes **print-ready docent + visitor artifacts** (Appendices) and a curated **external reference list**.

1) Core thesis (operational, falsifiable)

Two-ness is not a substance. It is the **invertibility class of a description** under a chosen **individuation basis**.

Operationally:

- “Two-ness disappears” **iff** we apply a **many-to-one map** (forgetting / aggregation) **and** we do **not** preserve a sufficient **audit channel** (receipt/provenance record) that would make the map invertible in practice.

Public hook: *Receipt kept $\rightarrow$ reversal possible. Receipt erased $\rightarrow$ reversal impossible (operationally).*

2) Minimal formal frame

Let token states live in a space X . Two labeled tokens live in X^2 . The swap group S_2 acts by $(x_1, x_2) \mapsto (x_2, x_1)$.

2.1 Forgetting maps

- **Label-forgetting (quotient by swaps):** $q: X^2 \rightarrow X^2/S_2$
- **Aggregation/coarse-graining:** $f: X^2/S_2 \rightarrow A$ where A is an aggregate description space (mixtures, totals, multisets, etc.).

A description preserves token identity exactly when the overall map is **injective** on the operational domain.

2.2 Audit channel (“receipt”) makes forgetting reversible

Introduce auxiliary information R (audit/provenance/permutation record) and consider

$\$(x_1, x_2) \mapsto (q(x_1, x_2), R). \ \$$

If R is sufficient, the augmented map can be injective even when q and f are not.

2.3 Information–thermodynamics link (what we do and do *not* claim)

- If you **erase** a record, the minimal dissipation scales with the **entropy you actually destroy**.
- If you **keep** correlations/side information, you can make processes reversible **up to** the moment you later reset/standardize that side memory.

Important nuance for the program:

- **Tier 0 (public):** show *qualitative* energy/erasure effects.
 - **Tier 2 (research):** only claim near- $kT \ln 2$ saturation with a physical 1-bit memory apparatus designed for that regime.
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3) Audience tiers (explicit)

Tier 0 — Museum/public (5–10 min)

Goal: felt understanding of basis dependence + the receipt toggle.

- Primary: **Clinic D** (basis swap + audit toggle)
- Optional add-ons: **Clinic A-lite** (energy vs “erase”), short HOM dip video loop
- Output: live tally/dashboard + take-home card

Tier 1 — Classroom / undergrad lab (60–120 min)

Goal: operational definitions + measurement + uncertainty.

- Core: **Clinic D as an experiment** (real dataset)
- Anchor physics (choose one):
- **Clinic C (HOM)** if optics lab exists
- **Clinic B** if you can outsource isotope assay
- Output: worksheet + analysis template

Tier 2 — Research-grade

Goal: publishable physics claims and extensions.

- **Clinic C (HOM)** as the cleanest controllable toggle
 - **Clinic A2** (physical 1-bit memory) for Landauer saturation and/or conditional-erasure extensions
 - Optional: formal “audit register” protocols (side information / conditional erasure)
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4) The clinics

Clinic D — Basis-swap + cognitive dataset (the engine)

Narrative: “How many things are here?” The answer changes when we change the bookkeeping rules.

Materials:

- Two distinct tokens (beads/coins) with removable labels
- Opaque capsule/box
- Sealed **Audit Envelope** (receipt card)
- Optional: tablet or paper tally

Conditions (within-subject; counterbalanced):

1. **Label basis:** A/B visible
2. **Type basis:** labels hidden; tokens identical
3. **Aggregate basis:** tokens placed into opaque capsule; treated as one object
4. **Audit toggle:** audit exists vs no audit (after aggregation)

Primary measures:

- Count judgment category: {“2 tokens”, “1 type (2 tokens)”, “1 object”, other}
- Confidence (0–10)
- Optional: response time, 1-sentence explanation

Core predictions:

- P1: Label basis increases “2 tokens” responses
- P2: Aggregate basis increases “1 object” responses
- P3 (key): Audit-present increases **reconstructibility** judgments *without necessarily changing perceptual oneness*

Pass/fail (Tier 0): observable distribution shift across conditions.

Clinic A — Records → Thermodynamics (two-tier)

A-lite (Tier 0/1): “Erasing the receipt costs something”

Goal: show that irreversible reset has an energetic cost in real devices (qualitative).

- Measure power draw during “store vs reset” operations on a microcontroller or simple memory module.
- Compare “overwrite/reset” vs “keep/append-only audit log.”

Important: do **not** claim saturation at $kT \ln 2$ in this tier.

A2 (Tier 2): Research-grade Landauer platform

Goal: measure dissipation near the Landauer bound using an apparatus built for that regime (e.g., physical double-well memory).

- Investigate scaling with erased entropy (including permutation-code entropy under stated distributions).
 - Optional extension: conditional erasure with side information (audit register).
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Clinic B — Forensic isotope melt-merge (material provenance)

Narrative: “Melt two into one—did two-ness vanish? What invariants survive?”

Core idea: compositional invariants (e.g., isotopic ratio) can survive aggregation; token-level trajectories typically do not, absent a provenance channel.

Tier guidance:

- Tier 0: describe conceptually (no melting on-site)
- Tier 1: controlled demo with outsourced assay + strong safety constraints

Safety: do not use legal tender; follow PPE and fume extraction; use approved materials.

Clinic C — Quantum indistinguishability (HOM “receipt toggle”)

Narrative: “Two photons: mark them and interference vanishes; unmark them and it returns.”

Control knob: distinguishability via polarization/path delay/spectral filtering.

Pass/fail: statistically significant visibility change: high when unmarked, low when marked, high again when unmarked.

5) Falsifiability wall statement (museum-safe)

If a receipt/audit record is preserved, ‘loss of two-ness’ is only apparent and is reconstructible. If the receipt is erased, reversibility fails operationally.

6) Fastest path to validation (momentum-first)

Immediate (run today)

- Deploy **Clinic D** with the docent script + visitor worksheet.
- Collect $N \approx 100\text{--}300$ responses at an event.
- Display a simple live tally dashboard.

Next (anchor physics)

- Add **Clinic C** if an optics lab exists (best quantitative toggle).
- Add **Clinic B** if you prefer a material-science anchor and can outsource isotope assay.

Research pathway

- Partner for **A2** (physical 1-bit memory) to claim Landauer saturation credibly.

7) Print-ready artifacts (already generated)

These are available as PDFs in the workspace:

- **Docent show card (Letter, 1 page):** `twoness_docent_show_card_letter.pdf`
- **Visitor worksheet (Letter, 2 pages):** `twoness_visitor_worksheet_letter.pdf`
- **Combined packet (Letter, 3 pages):** `twoness_clinics_packet_letter.pdf`

Appendix A — Docent Show Card (1 page)

TWO-NESS CLINICS: The Receipt Test

Core Question: Can you tell them apart later? Only if you kept the receipt. **Time:** 7 min | **Audience:** Museum/Public | **Tier:** 0

MATERIALS

- Two distinct tokens (e.g., beads, marked coins)
- Opaque capsule or small box
- “Audit Envelope” (sealed card showing Token A → Left, Token B → Right)
- Tablet/Post-it for live response tally (optional)

SCRIPT & FLOW (☐ = PASS/FAIL CHECK)

0. HOOK (30 sec)

"HOW MANY THINGS ARE HERE? Let's find out—and watch how the answer changes when we change the rules for keeping track."

Show two labeled tokens (A & B). Ask: "How many things?"

1. REMOVE THE RECEIPT (60 sec)

Hide/remove labels so tokens look identical. Ask again.

PASS: ☐ Audience count-judgment shifts (toward "one type/multiplicity").

2. AGGREGATE (60 sec)

Place both tokens into opaque capsule. Ask again.

PASS: ☐ Majority reports "one object."

3. THE AUDIT TOGGLE — THE KEY (90 sec)

Hold up sealed Audit Envelope. Ask whether it changes what *is* or what we can *know*.

CRITICAL PASS BOX:

- ☐ With audit: reconstruction judged possible
- ☐ Without audit: reconstruction judged impossible

4. LINK TO PHYSICS (60 sec)

"Erasing a physical receipt has a minimum energy cost per bit; keeping it in a side-channel avoids paying that cost *until you erase it later*."

Optional: show a 10s HOM dip toggle video.

5. HANDOFF (30 sec)

Invite participants to fill the worksheet / scan QR for live results.

Appendix B — Visitor Lab Worksheet (2 pages)

TWO-NESS CLINICS: Data Sheet

Prompt: Can you tell them apart later? Only if you kept the receipt.

PART 1: COUNT JUDGMENTS

For each scenario, circle ONE response + confidence (0–10).

A — Label basis: [2] [1 type (2 tokens)] [1 object] Other: ____ | Confidence: ____

B — Type basis: [2] [1 type (2 tokens)] [1 object] Other: ____ | Confidence: ____

C — Aggregate basis: [2] [1 type (2 tokens)] [1 object] Other: ____ | Confidence: ____

PART 2: THE AUDIT TOGGLE

Case 1: Audit exists → reconstruct which was which? [YES] [NO] Why (1 sentence):

Case 2: No audit → reconstruct which was which? [YES] [NO] Why (1 sentence):

PART 3: PREDICTION CHECK

1. If audit is kept, “loss” is: [REAL] [APPARENT]
2. If audit is erased, “loss” is: [REVERSIBLE] [IRREVERSIBLE] (operationally)

PART 4 (optional): HOM station

Record visibility and pass/fail.

External references (starter bibliography)

(Listed by theme; use DOI where available.)

Information → thermodynamics / Landauer

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Quantum indistinguishability / Hong–Ou–Mandel

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